

## Accepted Manuscript

This is an Accepted Manuscript of an article published by Taylor & Francis in *Research in Science & Technological Education* on 2 September 2026. The Version of Record is available online at: <https://doi.org/10.1080/02635143.2026.2727576>

## Gender and STEM: academic over-selection and stratified university entry

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## Abstract

Gender disparities in STEM participation remain one of the most persistent forms of horizontal segregation in higher education. While prior research has predominantly emphasised individual-level determinants such as interests, self-efficacy, or vocational orientations, less attention has been devoted to how gendered participation is embedded within structurally differentiated entry contexts. This study adopts an intersectional perspective to examine how educational trajectories, academic positioning, social origin, and life-course conditions combine into structured configurations at the point of university entry. Using administrative register data on a cohort of entrants to the Catalan Higher Education System in 2024–25 (N = 36,075), we apply Multiple Correspondence Analysis followed by cluster analysis to identify the principal dimensions structuring access to higher education. Degree subfields are projected as illustrative variables to assess how STEM disciplines are positioned within this entry structure. Results reveal a highly stratified entry space organised along a continuum contrasting early academic trajectories with vocational pathways. Women's participation in technical, experimental and biological STEM subfields is concentrated in the most academically advantaged profiles, whereas men access STEM from a broader range of entry contexts, indicating a pattern of academic over-selection in women's access to these fields.

**Keywords:** STEM fields; gender inequalities; horizontal segregation; degree choice; Multiple Correspondence Analysis

## **STEM career choice: an ongoing challenge**

STEM disciplines are widely regarded as essential to human development and are consistently associated with favourable labour-market outcomes, including low unemployment rates and strong economic prospects. Nevertheless, many countries have found that efforts to strengthen STEM education have not yielded the expected results. Consequently, the factors influencing students' decisions to pursue STEM careers have attracted increasing scholarly attention in recent years, with the overarching objective of reversing the current enrolment trend in the field (López, Simó, & Marco, 2023).

STEM career choices are widely recognised as complex phenomena shaped by the interaction of multiple factors, as documented across a substantial body of literature. Research has examined the role of attitudes (Osborne, Simon, & Collins, 2003), identity (Dou & Cian, 2022) and broader theories of educational and career choice, including Expectancy-Value Theory (EVT), Social Cognitive Career Theory (SCCT), RIASEC personality theory, and other developmental frameworks, to identify the most influential determinants of STEM participation. Over time, understanding of STEM career choice has evolved to reflect this growing theoretical and empirical diversity.

In a recent literature review, Zhou and Shirazi (2025) identify several key factors influencing young people's STEM career choices in post-2010 educational contexts, including self-motivation (encompassing interest and enjoyment and social utility motivation), social persuasion, self-efficacy, personal utility (such as financial security and job stability), and positive experiences of STEM. While the authors emphasise the need for future research to explore the intersectionality of these factors, we argue that STEM participation must also be understood in relation to the social dimensions structuring access to higher education.

A substantial body of sociological research has shown that students' choices are embedded in structurally differentiated opportunity contexts, where factors such as family educational background, prior educational trajectories and stratified access routes, academic positioning within the educational system, and life-course conditions such as age and employment status shape the range of options perceived as accessible (Boudon, 1974; Breen & Goldthorpe, 1997). Analysing how these social, educational and life-course dimensions combine provides a more accurate account of differentiated entry into STEM fields than examining each factor in isolation. Indeed, although gender remains a fundamental axis in the analysis of STEM inequalities, educational trajectories are conditioned by multiple dimensions that interact with one another. In this sense, a growing body of research advocates for an intersectional perspective (Gross, Gottburgsen, & Phoenix, 2016; Nichols & Stahl, 2019) to capture how gender combines with other factors such as social class, ethnicity or age to configure educational opportunities.

### *Gender disparities within STEM fields*

Beyond general explanations of STEM career choice, a major strand of this literature examines the trajectories and decisions of underrepresented students (Wong, Chiu, Murray, & Horsburgh, 2022). A substantial body of research has documented persistent gender disparities in participation across STEM fields, particularly in highly technical and object-oriented domains such as engineering, computer science and physics (Cheryan, Ziegler, Montoya, & Jiang, 2017; Su & Rounds, 2015; Wong et al., 2022). This literature shows that gender differences in STEM

participation are not evenly distributed across fields: women tend to be more strongly represented in life-science, health-related and people-oriented areas, while remaining underrepresented in many technical and engineering-related subfields. Existing explanations have emphasised gendered vocational interests, communal and agentic goal orientations, self-efficacy, sense of belonging, stereotypes about who “fits” in STEM, and field-specific disciplinary cultures (Cheryan et al., 2017; Master, Cheryan, & Meltzoff, 2016; Su & Rounds, 2015). Building on these findings, this study examines whether gendered participation across STEM subfields is also embedded in differentiated university-entry profiles, shaped by prior educational trajectories, academic positioning, family educational background and life-course conditions.

One influential line of research has emphasised the role of vocational interests and values in explaining broad gendered patterns of educational participation. Studies indicate that gender differences in educational participation are often linked to divergent value orientations, with women more likely to prioritise communal and socially oriented goals and men more frequently expressing agentic or competitive orientations. This difference in values has been proposed as a primary explanation for the underrepresentation of women in highly technical fields, contrasting with their greater presence in disciplines such as social sciences, health, or life sciences (Diekmann, Brown, Johnston, & Clark, 2010; Su, Rounds, & Armstrong, 2009). Importantly, these differences do not emerge in isolation, but are shaped by broader socialisation processes and cultural expectations regarding appropriate gender roles.

Beyond vocational interests, sociocultural explanations have highlighted the role of learning environments, disciplinary cultures and gendered norms within STEM fields. Research has shown that highly masculinised fields often convey implicit signals about who belongs, through classroom dynamics, peer cultures, stereotypes about competence and representations of the “typical” student or professional. Such environments may undermine women’s sense of belonging and identification with the field, even when their academic performance is comparable to that of men. The scarcity of female role models and the persistence of stereotypes portraying STEM professionals, particularly in computer science and physics, as male, socially isolated or exclusively focused on technical tasks further contribute to these dynamics (Cheryan et al., 2017).

In addition to educational experiences and cultural factors, recent literature highlights the role of anticipated labour-market conditions in shaping gendered patterns of participation in STEM. Studies show that women are less likely to enter fields associated with long working hours, limited temporal flexibility and high penalties for career interruptions, and are more likely to favour trajectories linked to more predictable and regulated professional environments (England, Bearak, Budig, & Hodges, 2016; Goldin & Katz, 2016).

Although these mechanisms operate later in the life course, they are likely to influence educational choices indirectly, through prospective expectations about future employment conditions and the compatibility of different careers with family life. These anticipatory considerations are particularly salient in contexts where certain STEM-related professions are associated with demanding work schedules and strong norms of continuous availability. By contrast, fields linked to more regulated labour markets or clearer career structures often within the public sector may be perceived as offering greater protection against career penalties related to care responsibilities. These dynamics suggest that gendered participation in STEM is also

shaped by broader expectations about future employment conditions and work–life compatibility (Barone & Assirelli, 2020; Goldin, 2014; Mora, García-Aracil, & Vila, 2007).

Although STEM fields are often treated as a single analytical category, a growing body of research has highlighted substantial heterogeneity in terms of disciplinary cultures, practices and patterns of student participation (Cheryan et al., 2017; Su & Rounds, 2015). Gender disparities vary markedly across STEM subfields, indicating that aggregate analyses risk obscuring meaningful internal differences. Women remain strongly underrepresented in areas such as engineering, computer science and physics, while their participation is substantially higher, and in some cases dominant, in fields such as biology, chemistry and parts of the life sciences (Cheryan et al., 2017). Architecture programmes, in turn, show substantial cross-national variation but often relatively high female participation, with women representing close to or more than half of enrolled students in several contexts (Charles & Bradley, 2009).

Importantly, considerable variation also exists within engineering itself. Biomedical engineering frequently approaches gender parity, and chemical engineering tends to include a substantially higher proportion of women than many other engineering branches. By contrast, mechanical, electrical and computer engineering remain among the most male-dominated specializations (Godwin, Potvin, Hazari, & Lock, 2016).

These contrasting patterns challenge approaches that treat STEM as a homogeneous domain, suggesting instead that gender disparities are shaped by field-specific characteristics and differentiated institutional contexts. Building on research on vocational interests, Su and Rounds (2015) show that gender differences across STEM subfields align with a continuum reflecting the degree to which disciplines are oriented towards working with ‘things’ (objects, systems, technical artefacts) versus working with ‘people’. Subfields with a strong object-oriented profile, such as engineering and computer science, tend to display the largest gender gaps, whereas those more closely connected to communal or people-oriented goals, such as biology, exhibit more balanced participation.

However, internal differentiation within STEM cannot be reduced to interest profiles alone. Studies have pointed to substantial differences across STEM subfields in terms of learning environments, peer cultures, and norms of belonging (Cheryan et al., 2017; Godwin et al., 2016). The gendered signals and ‘typical’ representations identified above do not manifest uniformly across STEM disciplines (Master et al., 2016). Masculinised environments reproduce dynamics that differentially affect students’ sense of fit and identity. As a result, STEM subfields differ not only in their gender composition, but also in the types of students they attract, the institutional entry contexts with which they are associated, and the estimated risks of entering them, including the potential for identity mismatch in highly masculinised environments.

Taken together, this literature suggests that STEM should not be conceptualised as a homogeneous space, but rather as a constellation of fields characterised by distinct academic demands, cultural norms and patterns of participation (Su & Rounds, 2015; Veldman, Van Laar, Thoman, & Van Soom, 2021). These internal differences raise the question of whether STEM subfields also differ systematically in the social and educational profiles of students who enter them, a question that requires moving beyond gender alone and considering how multiple

dimensions of students' trajectories intersect across fields (Gross et al., 2016; Nichols & Stahl, 2019).

### *Stratified university entry and intersectional configurations*

International research has shown that gender segregation by field of study is structured by institutional contexts and varies across educational systems (Charles & Bradley, 2009), while educational inequalities operate not only vertically but also horizontally across fields of study (Reimer & Pollak, 2010). This suggests that gendered participation in STEM must be understood in relation to stratified educational pathways and differentiated opportunity structures.

Moreover, STEM career progression has often been conceptualised through the leaky pipeline metaphor, which portrays STEM careers as linear and predetermined pathways. This perspective oversimplifies students' decision-making processes, overlooks contextual influences, and neglects the existence of alternative and non-linear routes into STEM, thereby obscuring the substantial number of individuals who ultimately establish careers in STEM fields despite discontinuous trajectories (Li, Kirk, & Oseguera, 2025). While research on STEM careers has extensively examined high-impact practices for student engagement and early exposure, comparatively little attention has been paid to the experiences of students navigating nontraditional pathways (Smith, 2025). While the definition of non-traditional students is under discussion (Brozina, Johri, & Chew, 2024), what some authors define as *post-traditional* learners have not been extensively included in studies regarding STEM career choices (Secules, Berhane, Long, Caringella, & Pinto, 2021).

These dynamics may become especially visible in contexts of heightened structural uncertainty. Evidence from university entrant cohorts during periods of economic crisis or tuition fee increases indicates that gender differences in degree choice tend to widen under conditions of greater constraint (Troiano, Torrents, Sánchez-Gelabert, & Daza, 2017). Rather than reflecting isolated individual preferences, these divergences point to gendered responses embedded within broader institutional and economic contexts.

Another key dimension shaping these differentiated trajectories is the diversification of access routes. The traditional linear pathway, from general secondary education (Bachillerato) directly to university, now coexists with alternative routes such as upper secondary Vocational Education and Training (VET). Students entering through vocational pathways often present distinct academic and social profiles compared to those from the academic track (Daza, Troiano, & Elias, 2019), with a higher presence of men and older students in non-traditional itineraries, particularly those leading to technical fields (Sánchez-Gelabert, 2020).

From a life-course perspective, age and employment status are not merely supplementary variables but structuring dimensions of university entry. The coexistence of study and paid work, as well as delayed entry into higher education, generate differentiated contexts of opportunity (Sánchez-Gelabert, 2024). Crucially, these contexts are themselves gendered. In highly demanding, face-to-face university environments, rigid schedules and intensive academic dedication may align differently with male and female trajectories, particularly where care responsibilities or external constraints are unevenly distributed (Triventi, 2014).

Taken together, this literature identifies three limitations that motivate the present study. First, STEM participation is often analysed either at an aggregate level or through broad gender comparisons, despite substantial heterogeneity across subfields. Second, explanations centred on interests, belonging or disciplinary cultures pay less attention to how students are sorted into fields through stratified university-entry contexts. Third, non-linear and vocational access routes remain insufficiently integrated into analyses of gendered STEM participation. Addressing these limitations, this study examines how educational trajectories, academic positioning, social origin and life-course conditions combine into structured configurations at the point of university entry, and how these configurations map onto access to STEM and STEM-adjacent subfields.

## Method

### Objective

The aim of this article is to provide an empirical exploration of patterns of entry into university fields of study, with a particular focus on the internal differentiation of STEM fields. To this end, the analysis identifies the main dimensions structuring access to higher education by jointly considering gender, prior educational trajectories, academic performance, family educational background, and indicators of students' life circumstances and stated motivations. The analysis approaches university entry as a key mechanism through which inequalities of opportunity are structured across fields of study.

More specifically, the study addresses three research questions:

- What are the main social, academic and educational-trajectory dimensions structuring entry into university fields of study?
- How do these dimensions combine into distinct student profiles at the point of university entry?
- How are different degree subfields, and particularly STEM subfields, positioned within this multidimensional space of university entry?

### Analytical strategy

From an analytical perspective, the study adopts Multiple Correspondence Analysis (MCA), an exploratory technique for identifying associations among categorical variables (Le Roux & Rouanet, 2010). MCA represents individuals and categories in a multidimensional space, where proximity indicates similar response profiles. This approach is appropriate because the study seeks to identify combinations of social, educational and life-course characteristics rather than estimate independent effects, and to locate degree subfields within the same factorial space.

The MCA was constructed using active variables, which contribute to the formation of the factorial axes. These active variables capture the main social, educational and life-course characteristics of students at the point of university entry. The resulting axes summarise the strongest associations among these categories. Categories that contribute strongly to an axis are used to interpret its substantive meaning, while categories with low contributions are not central to the interpretation. In addition, degree subfield and STEM classification were included as illustrative variables: they were projected onto the factorial space after the axes had been

constructed and therefore did not contribute to their formation. Their position should consequently be interpreted as an association with the profiles defined by the active variables.

Cluster analysis was then applied to the individual MCA coordinates to summarise the heterogeneity of student profiles and relate them to fields of study. Hierarchical clustering with Ward's minimum variance method was used, grouping individuals by minimising within-cluster variance. The final number of clusters was selected by combining the hierarchical tree, substantive interpretability and empirical validity criteria. Partitions from six to twelve clusters were compared using the inter-cluster inertia ratio, marginal gains in explained inertia, the Calinski–Harabasz pseudo-F index, average silhouette coefficients and minimum cluster size. The eight-cluster solution was retained because it produced the highest Calinski–Harabasz value while maintaining relatively balanced cluster sizes and avoiding fragmentation in larger solutions.

## Data

The analysis draws on administrative register data from the Catalan University Pre-Enrolment Office (Oficina de Preinscripció Universitària, Generalitat de Catalunya). The study was based on secondary analysis of administrative microdata without direct personal identifiers. Access to the microdata was obtained through the formal administrative request procedure of the Government of Catalonia; the request was assessed and authorised by the competent office, after which the dataset was made available for research purposes. The dataset covers all applicants to undergraduate programmes offered by the 12 universities in the Catalan higher education system in the 2024–25 academic year ( $N = 57,159$ ). The file received for the study did not contain direct personal identifiers such as names, surnames or national identity numbers. No direct contact with students took place, and the study involved no intervention with participants. The authors had no access to directly identifying information, did not attempt to re-identify individuals, and analysed the data only in aggregate form.

The analytical sample is restricted to students who effectively enrolled at university and accessed higher education through the two predominant entry routes: the academic track (upper secondary education) and the professional track (upper-secondary vocational training). Applicants who did not ultimately enrol were excluded from the analysis ( $n = 13,213$ ; 23.1% of all applicants). To ensure comparability in admission grades and educational trajectories, applicants entering through alternative access routes (1.3%) and students with previous university enrolment (18.1%) were also excluded. These exclusions reflect the study's focus on first-time university entry rather than re-entry, programme change, second-degree enrolment or alternative access trajectories, which may involve different admission mechanisms, age profiles and educational histories. Finally, a very small number of records with incomplete or inconsistent information preventing classification into one of the degree subfields analysed were removed ( $n = 122$ ; 0.34% of the analytical sample). After applying these sample restrictions and data-cleaning procedures, the final sample comprises 36,075 first-time university entrants enrolled in 2024–25.

## Variables and measures

To construct the factorial space, the analysis relies on a set of active categorical variables capturing key dimensions of students' situations at the moment of university entry. These

include sociodemographic characteristics (age, nationality, gender), social origin (family educational background), social conditions (employment status), educational factors (access route and admission grade), and motivational factors (stated motives for degree choice). A detailed overview of variables and categories is provided in Table 1.

The resulting dimensions were subsequently used as input for a classification analysis aimed at identifying groups of students sharing similar configurations of characteristics, thereby capturing internal heterogeneity in university entry.

*Table 1. Active variables and categories*

| Variable                               | N      | %    |
|----------------------------------------|--------|------|
| Sex                                    |        |      |
| Male                                   | 15,319 | 42.5 |
| Female                                 | 20,756 | 57.5 |
| Nationality                            |        |      |
| Spanish                                | 33,047 | 91.6 |
| Non-Spanish                            | 3,028  | 8.4  |
| Employment status                      |        |      |
| Not working                            | 28,549 | 79.1 |
| Up to 15 hours/week                    | 2,958  | 8.2  |
| 15 or more hours/week                  | 4,568  | 12.7 |
| Access route                           |        |      |
| Academic track (Bachillerato)          | 29,877 | 82.8 |
| Professional track (Upper VET)         | 6,198  | 17.2 |
| Family educational level               |        |      |
| Compulsory education                   | 6,841  | 19.6 |
| Post-compulsory education              | 8,079  | 23.1 |
| University education                   | 20,016 | 57.3 |
| Age                                    |        |      |
| Up to 18                               | 25,924 | 71.9 |
| 19–20                                  | 6,464  | 17.9 |
| 21–24                                  | 3,014  | 8.4  |
| 25 or older                            | 673    | 1.9  |
| Motives for degree choice <sup>1</sup> |        |      |
| Interest                               | 20,412 | 56.6 |
| Career expectations                    | 10,891 | 30.2 |
| Pragmatic and logistical factors       | 969    | 2.7  |
| Admission cut-off mark                 | 2,067  | 5.7  |
| Other                                  | 1,736  | 4.8  |
| Admission grade (deciles)              |        |      |

<sup>1</sup> The original item on ‘Motives for degree choice’ included several residual categories. Responses referring to programme duration (0.5%), perceived difficulty (1.1%) and geographical proximity (1.1%) were grouped into a single category labelled “Pragmatic and logistical factors” (2.7%), capturing considerations related to study feasibility and conditions. This aggregation was applied to avoid sparse categories while preserving conceptual coherence.

|                              |        |       |
|------------------------------|--------|-------|
| 1st decile ( $\leq 7.08$ )   | 3,588  | 9.9   |
| 2nd decile (7.09–7.95)       | 3,599  | 10.0  |
| 3rd decile (7.96–8.66)       | 3,612  | 10.0  |
| 4th decile (8.67–9.31)       | 3,603  | 10.0  |
| 5th decile (9.32–9.89)       | 3,611  | 10.0  |
| 6th decile (9.90–10.42)      | 3,629  | 10.1  |
| 7th decile (10.43–10.96)     | 3,606  | 10.0  |
| 8th decile (10.97–11.55)     | 3,608  | 10.0  |
| 9th decile (11.56–12.30)     | 3,607  | 10.0  |
| 10th decile ( $\geq 12.31$ ) | 3,612  | 10.0  |
| Total                        | 36,075 | 100.0 |

In addition, degree subfield was included as the main illustrative variable. This variable comprises 23 subdisciplinary categories and covers all degree programmes in the analytical sample. Degree subfields are defined according to the official classification of degree programmes established by the Catalan University Quality Assurance Agency (AQU Catalunya) in its Catalogue of Degree Programmes. A secondary STEM/non-STEM aggregation was also used to facilitate comparison with prior literature on STEM participation. This aggregation is derived from the 23 subdisciplinary categories and follows the operational definition adopted in the study, which restricts STEM to scientific, technological, engineering and mathematical domains and treats health-related fields as a separate area. The distribution of illustrative variables is reported in Table 2.

This decision is consistent with broad field-of-study classifications and international statistical reporting, including ISCED-F (UNESCO Institute for Statistics, 2015), OECD (2025) and Eurostat (2024), where STEM comprises natural sciences, mathematics and statistics, information and communication technologies, and engineering, manufacturing and construction, while Health and Welfare constitutes a distinct field. This distinction is also reflected in the Spanish Higher Education Statistics (INE) and in the Catalan Quality Assurance Agency (AQU Catalunya) classification.

Both degree subfield and the STEM/non-STEM aggregation were included as illustrative variables. They were projected onto the factorial space after the axes had been constructed and therefore did not contribute to the formation of the dimensions. Consequently, the position of degree subfields in the factorial space should be interpreted as an association with the student profiles defined by the active variables, not as a determinant of the space itself.

*Table 2. Illustrative variables and categories*

| Subdiscipline                                    | N     | %    | STEM |
|--------------------------------------------------|-------|------|------|
| Economics, Business and Tourism                  | 4,717 | 13.1 | No   |
| Nursing and Healthcare                           | 4,449 | 12.3 | No   |
| Industrial Technologies                          | 3,543 | 9.8  | Yes  |
| Law, Labour Relations and Political Sciences     | 3,172 | 8.8  | No   |
| Information and Communication Technologies (ICT) | 2,281 | 6.3  | Yes  |
| Education                                        | 2,194 | 6.1  | No   |

|                                                  |        |       |     |
|--------------------------------------------------|--------|-------|-----|
| Medicine and Biomedical Sciences                 | 1,692  | 4.7   | No  |
| Biological and Earth Sciences                    | 1,445  | 4.0   | Yes |
| Joint Programmes (Social Sciences)               | 1,340  | 3.7   | No  |
| Communication and Documentation                  | 1,292  | 3.6   | No  |
| Languages and Literatures                        | 1,289  | 3.6   | No  |
| Psychology and Therapy                           | 1,162  | 3.2   | No  |
| Joint Programmes (Engineering)                   | 1,152  | 3.2   | Yes |
| Arts and Design                                  | 1,093  | 3.0   | No  |
| Philosophy and History                           | 1,077  | 3.0   | No  |
| Experimental Sciences and Mathematics            | 1,069  | 3.0   | Yes |
| Architecture, Construction and Civil Engineering | 986    | 2.7   | Yes |
| Social Intervention                              | 825    | 2.3   | No  |
| Agriculture, Forestry and Fisheries              | 636    | 1.8   | Yes |
| Joint Programmes                                 | 282    | 0.8   | Yes |
| Joint Programmes (Humanities)                    | 160    | 0.4   | No  |
| Joint Programmes (Health)                        | 150    | 0.4   | No  |
| Joint Programmes (Sciences)                      | 69     | 0.2   | Yes |
| Total                                            | 36,075 | 100.0 |     |

| STEM Field | N      | %     |
|------------|--------|-------|
| STEM       | 11,463 | 31.8  |
| Non-STEM   | 24,612 | 68.2  |
| Total      | 36,075 | 100.0 |

## Results

The results are organised around the three research questions: first, the main dimensions structuring university entry; second, the student profiles identified within this space; and third, the positioning of STEM and health-related subfields across these profiles.

### *Main dimensions structuring university entry*

As a first step in the analysis, the main dimensions structuring the space derived from the Multiple Correspondence Analysis (MCA) were identified<sup>2</sup>. These dimensions were selected on the basis of their eigenvalues, which indicate the amount of inertia (variance) accounted for by each factor, decreasing progressively across successive axes. Following the procedure proposed by Le Roux and Rouanet (2010), Benzécri's correction was applied to assess the relative importance of the factors and to determine their explanatory power. The retained dimensions provide a parsimonious representation of the relationships among the variables, allowing the visualisation and interpretation of how students' life circumstances and educational trajectories are jointly organised.

<sup>2</sup> Residual categories and missing values were treated as supplementary modalities and projected a posteriori, without contributing to the construction of the factorial axes.

Table 3. MCA with selection of active variable

| Factors | Eigenvalue | Corrected eigenvalue | % corrected inertia | % cumulative inertia |
|---------|------------|----------------------|---------------------|----------------------|
| 1       | 0.2777     | 0.023                | 96.0                | 96.0                 |
| 2       | 0.1563     | 0.001                | 4.0                 | 100.0                |

Table 3 reports the eigenvalues and the proportion of inertia explained by the factors retained in the analysis. After applying Benzécri's correction, the first factor accounts for the vast majority of the explained inertia (96%), while the second factor captures a much smaller share of the remaining variance (4%). This highly concentrated structure indicates that the differentiation among students is strongly hierarchical and organised along a single dominant dimension. Given the limited corrected inertia explained by this second axis, its interpretation should be treated with caution. The axis is used here only to refine the description of specific profiles and not as the main basis for the substantive conclusions of the article.

However, while the first axis summarises the principal stratification of students' educational and social conditions, the second axis remains analytically relevant. Despite its lower statistical weight, the second axis suggests a secondary contrast related to academic positioning and nationality, which help characterise specific minority profiles and differentiated entry contexts. Consequently, the interpretation focuses primarily on the first factor as the structural axis of university access, while the second factor is examined as a complementary dimension that refines the characterisation of specific subgroups.

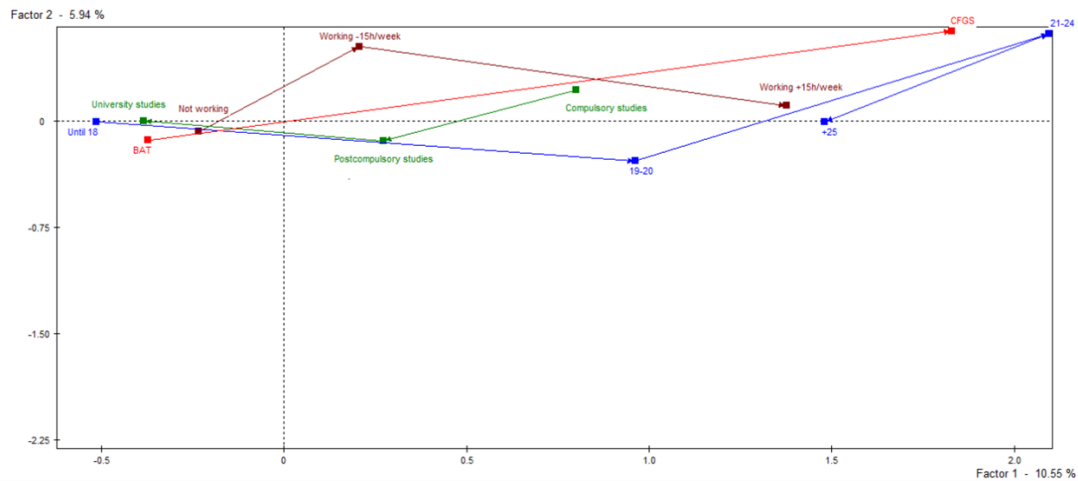
Table 4. The variables and modalities' contributions to the factors

| Variables Labels                                                          | Cont. | Modalities – negative coord. Label | Contr. | Modalities – positive coord. Label | Contr.   |
|---------------------------------------------------------------------------|-------|------------------------------------|--------|------------------------------------|----------|
| <i>Axis 1 – Life-course position and institutionalised access pathway</i> |       |                                    |        |                                    |          |
| Age                                                                       | 32.9  | Until 18                           | 8.7    | 19-20   21-24                      | 7.6 16.7 |
| Access route                                                              | 31.1  | Academic track                     | 5.2    | Professional track (Upper VET)     | 25.9     |
| Employment status                                                         | 12.9  |                                    |        | Working +15h/week                  | 10.8     |
| Family Education                                                          | 10.0  | University                         | 3.8    | Compulsory                         | 5.5      |
| <i>Axis 2 – Academic and institutional constraint gradient</i>            |       |                                    |        |                                    |          |
| Nationality                                                               | 30.6  | Non-Spanish                        | 28.1   |                                    |          |
| Admission grade                                                           | 25.2  | First decile ( $\leq 7.08$ )       | 16.8   |                                    |          |
| Motives for choice                                                        | 18.0  | Pragmatic and logistical factors   | 11.5   | Interest                           | 3.4      |
| Sex                                                                       | 12.4  | Male                               | 7.3    | Female                             | 5.2      |
| Access route                                                              | 6.7   |                                    |        | Professional track (Upper VET)     | 5.2      |

The interpretation of the factorial axes is based on the relative contributions of the active variables and their most influential categories (Table 4). Factor 1 is primarily structured by age (32.9%) and access route (31.1%), followed by employment status (12.9%) and family educational background (10.0%).

On the negative side of the axis, the most contributive categories correspond to younger students (up to 18 years old), entry through the general academic track (*Bachillerato*), and a family background with university-educated parents. Conversely, the positive side is dominated by older students (19–20 and 21–24), entry through upper secondary vocational education (Upper VET), intensive participation in paid work (>15 hours per week), and lower family educational levels (Figure 1).

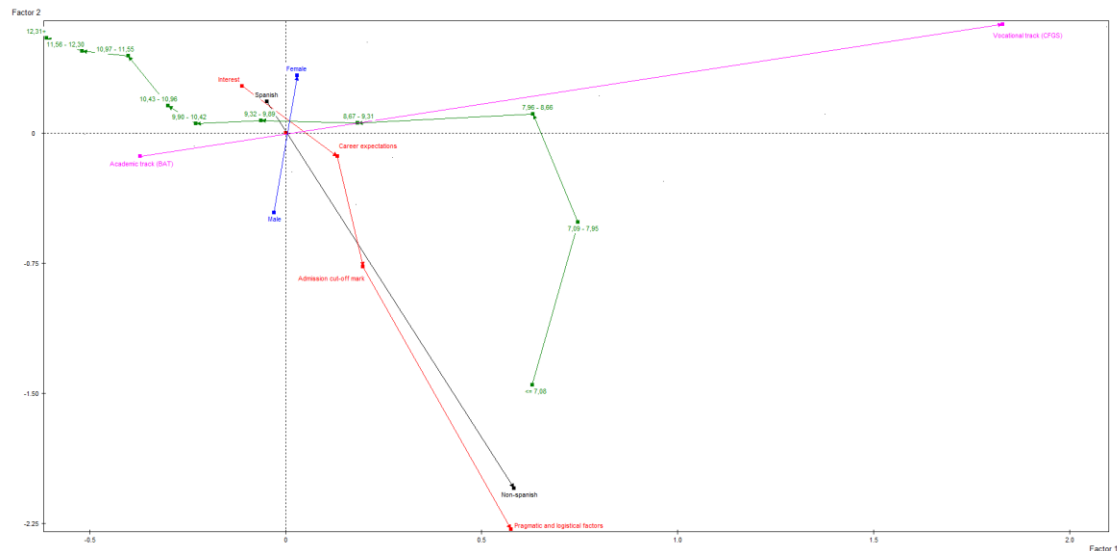
Figure 1. Projection of the variables and modalities that contribute most to factor 1



Taken together, this factor captures a sharp contrast between an early, academically linear entry into university typical of students from highly educated families, and a later, vocationally mediated entry associated with employment and lower parental educational attainment. The first axis can therefore be interpreted as the structural continuum of institutionalised access, distinguishing between traditional (linear) and non-traditional (delayed/vocational) trajectories.

Factor 2 is mainly defined by nationality (30.6%) and admission grade (25.2%), followed by stated motives for degree choice (18.0%) and sex (12.4%). The dominant contribution of nationality and academic credentials suggests that this axis captures a secondary contrast in the degree of institutional constraints and academic resources shaping access.

Figure 2. Projection of the variables and modalities that contribute most to factor 2



As shown in Figure 2, the negative side of the axis is characterised by non-Spanish nationality, lower admission grades (first decile,  $\leq 7.08$ ), and a stronger emphasis on pragmatic and logistical factors, such as duration or proximity, when choosing a degree. Interestingly, the male modality also contributes to this pole. These elements suggest entry contexts characterised by greater constraints, where degree choice is more strongly shaped by concerns about admission feasibility and logistical barriers rather than pure preference.

By contrast, the positive side of the axis is associated with interest as the primary motivation and, to a lesser extent, a greater presence of female students. This pole suggests entry contexts in which programme choice appears less constrained by pragmatic or institutional considerations. Although non-Spanish students represent less than 10% of total enrolment in the Catalan university system, their association with lower admission grades and feasibility-oriented motivations contributes to the formation of this secondary dimension. Given the limited corrected inertia explained by Axis 2, this pattern should be interpreted cautiously.

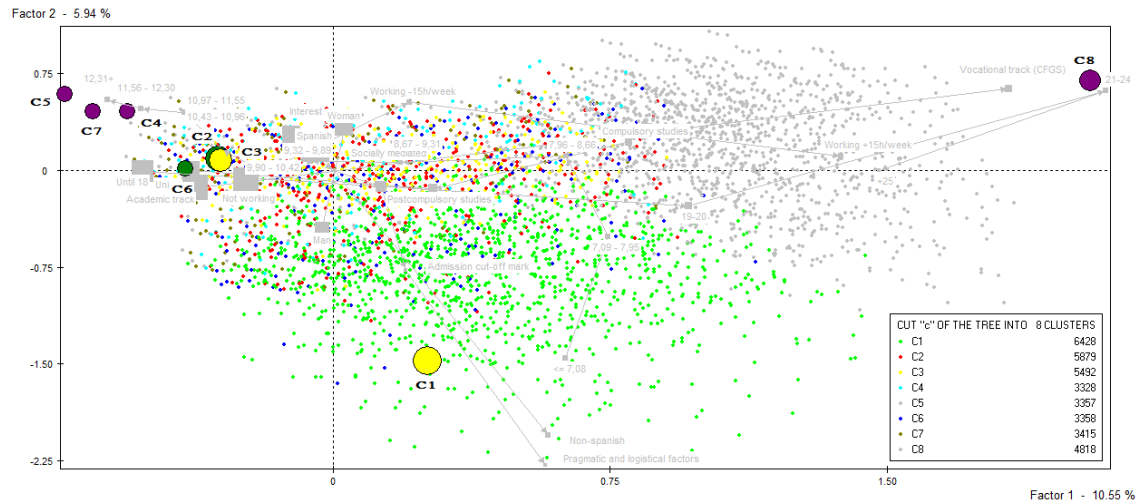
In summary, while the first axis describes the type of trajectory (linear vs. vocational), this second factor captures a secondary contrast associated with admission grade, nationality and pragmatic motivations. It is interpreted as suggesting a secondary contrast between students who navigate the system with high academic capital and intrinsic motivation and those associated with lower grades, non-Spanish nationality or pragmatic limitations.

### *Student profiles in the multidimensional space*

Cluster analysis identified eight empirical profiles of university entry within the factorial space (Figure 3). These profiles are characterised by overrepresented combinations of access route, age, admission grade, family educational background, employment status, gender composition and motivational orientation, and their main characteristics are summarised in Table 5.

Based on their position and proximity within the factorial space, the eight clusters reveal substantial internal heterogeneity. On the one hand, a group of profiles located in the upper-left quadrant (Clusters 2 to 7) is characterised by early, academically linear trajectories, entry through the academic track, full-time dedication to studies, and higher parental educational attainment. Together, these clusters account for approximately 68.8% of all university entrants.

Figure 3. Projection of clusters and active modalities on the first two MCA dimensions



Note: Circle size (radius) is proportional to each cluster's relative weight in the sample. Purple clusters indicate profiles with a higher female representation; yellow clusters indicate profiles with a higher male representation; and green clusters represent gender-balanced or mixed profiles.

Within this set, two subgroups can be distinguished according to gender composition, academic performance, and motivational orientation. Clusters 4, 5 and 7 are predominantly composed of female students with high academic performance (8th, 9th and 10th deciles of admission grades) and intrinsic or vocational motivations. In contrast, Clusters 2, 3 and 6 display greater heterogeneity. Clusters 2 and 6 are mixed in terms of gender composition, while Cluster 3 is more strongly associated with male students. A similar pattern emerges with respect to admission grades: Clusters 3 and 6 are concentrated in the middle deciles (4th to 6th), whereas Cluster 2 shows greater internal variability.

Table 5. Characteristics of the clusters

| Cluster (%) | Sex    | Access route | Admission grade | Family educational background | Employment status | Specific features        |
|-------------|--------|--------------|-----------------|-------------------------------|-------------------|--------------------------|
| C1 (17.8%)  | Male   | Academic     | 1st–2nd decile  | Post-compulsory               | No                | Non-Spanish; constraints |
| C2 (16.3%)  | Mixed  | Academic     | Heterogeneous   | University                    | <15 h/week        | -                        |
| C3 (15.2%)  | Male   | Academic     | 4th–5th decile  | Post-compulsory               | No                | Intrinsic mot.           |
| C4 (9.2%)   | Female | Academic     | 8th decile      | University                    | No                | Intrinsic mot.           |
| C5 (9.3%)   | Female | Academic     | 10th decile     | University                    | No                | Intrinsic mot.           |
| C6 (9.3%)   | Mixed  | Academic     | 6th decile      | University                    | No                | -                        |
| C7 (9.5%)   | Female | Academic     | 9th decile      | University                    | No                | Intrinsic mot.           |
| C8 (13.4%)  | Female | Upper VET    | 1st–4th decile  | Comp. / Post-comp.            | Yes               | Instrumental mot.        |

At the opposite extreme of the factorial space lies Cluster 8 (13.4% of entrants), characterised by non-linear trajectories, vocational access routes, older age profiles, and employment commitments. This cluster is predominantly composed of female students with instrumental motivations and admission grades concentrated in the lower part of the distribution (1st to 4th deciles). These characteristics indicate entry contexts marked by greater structural constraints and more limited academic positioning.

In an intermediate position within the space, Clusters 1 and 2 display distinctive configurations. Cluster 1 (17.8%), the largest group, is composed mainly of male students entering through the academic track with low admission grades (1st and 2nd deciles) and full-time dedication to studies. This cluster also shows a higher presence of non-Spanish students and motivations more closely associated with adaptive or constraint-driven considerations. Cluster 2 (16.3%) is defined by entry through the academic track, heterogeneous admission grades, university-educated family background, part-time employment (less than 15 hours per week), and a heterogeneous gender composition.

Overall, these results indicate that the presence of female students is unevenly distributed across the factorial space. Women are concentrated, on the one hand, in academically highly selective profiles (Clusters 4, 5 and 7) and, on the other, in more constrained and non-traditional entry contexts (Cluster 8), with clearly differentiated conditions between these two poles. Men, by contrast, are more visible in lower and intermediate academic-track positions, notably Clusters 1 and 3, and remain present in academic-track entry even at lower levels of academic performance, a configuration less frequently observed among female students.

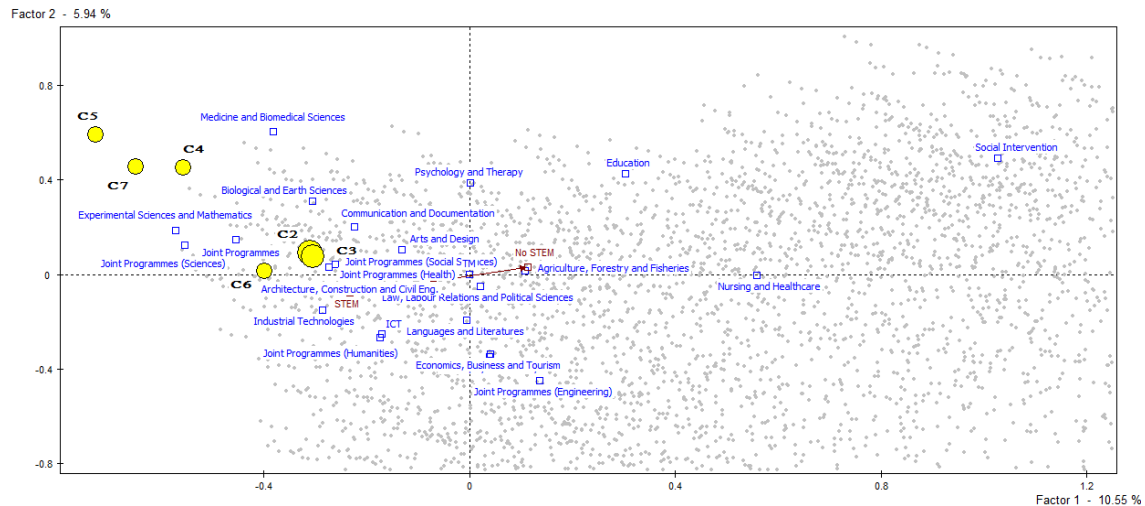
This indicates that gender does not structure the factorial space as a primary axis, but becomes highly salient within specific entry conditions. In this sense, gendered patterns of participation appear embedded within structurally differentiated entry contexts rather than operating as an autonomous organising principle. Projecting subfields of study onto this space allows us to examine how these distinct student profiles map onto access to STEM disciplines.

#### *STEM enrolment and student profiles*

The projection of degree subdisciplines onto the factorial space provides a substantive interpretation of how subfields of study are embedded within the differentiated contexts of university entry identified by the MCA. Although these subdisciplines do not contribute to the construction of the axes, their position within the space allows for a systematic assessment of the types of entry contexts with which different fields are associated. The joint projection of clusters and degree subfields is shown in Figure 4.

Along the first axis, a clear and structured contrast emerges between STEM and non-STEM fields. STEM degrees are located on the negative side of the axis, indicating that STEM enrolment is predominantly associated with early, academically mediated entry trajectories. This positioning aligns STEM fields with contexts characterised by younger age at entry, access through the academic track (Bachillerato), full-time dedication to studies, and higher parental educational attainment.

*Figure 4. Zoomed projection of clusters and degree subfields onto the factorial space*



Note: The figure shows an enlarged view of the region where degree subfields are concentrated in order to facilitate interpretation; clusters located outside this range are not displayed.

Within STEM, this pattern is particularly pronounced for Experimental Sciences, Industrial Technologies, and Biological Sciences, all of which display strongly negative coordinates on the first axis. It is also noteworthy that degrees in Medicine and Biomedical Sciences, although not classified as STEM according to the operational classification adopted in this study, are positioned in the same negative region of the axis, closely aligned with several STEM disciplines in terms of entry context.

By contrast, non-STEM fields are positioned toward the central and positive side of the first factor, corresponding to later or more constrained entry contexts. Subdisciplines such as Education, Social Intervention, and especially Nursing and Healthcare degrees show the strongest positive coordinates. These fields are systematically associated with older age at entry, vocational access routes, a higher likelihood of combining studies with paid work, and lower parental educational attainment.

These results also highlight the internal heterogeneity of health-related fields. Although health-related degrees are not classified as STEM in the operational definition adopted in this study, their position in the factorial space is not uniform. Medicine and Biomedical Sciences are located close to several STEM subfields, sharing an entry context characterised by academic-track access, high admission grades and higher parental educational attainment. By contrast, Nursing and Healthcare is positioned closer to Education and Social Intervention, and is associated with older age at entry, vocational access routes, employment commitments and lower parental educational attainment.

The projection of subdisciplines also allows us to relate the student profiles identified through the cluster analysis to specific fields of study, based on their proximity in the factorial space and the relative weight of each subdiscipline within clusters (Table 6). When examining the distribution of STEM across student profiles, a clear gendered pattern emerges.

Among female students, enrolment in technical, experimental and biological STEM subfields is highly concentrated in clusters characterised by strong academic positioning and favourable structural conditions (Clusters 4, 5 and 7). These clusters combine high admission grades, academic-track access, full-time dedication to studies and higher family educational background. In Cluster 4, STEM participation includes Biological Sciences, Industrial

Technologies, ICT and Architecture and Construction. Cluster 5 combines STEM subfields such as Experimental Sciences and Industrial Technologies with Medicine and Biomedical Sciences. Cluster 7 is strongly associated with Biological Sciences, Experimental Sciences and Industrial Technologies, while Medicine and Biomedical Sciences is also overrepresented. Although Medicine and Biomedical Sciences is not classified as STEM in the operational definition adopted here, its presence in these clusters indicates proximity in terms of entry-profile characteristics, particularly high admission grades and academic-track access. Overall, these results show that women's participation in technical STEM subfields, including areas with particularly low female representation such as Industrial Technologies and ICT, is embedded within highly selective academic trajectories.

*Table 6. Main disciplinary associations by cluster*

| Cluster | Sex composition | STEM orientation | Main associated subdisciplines (%)                                                                                                                                                                                                              |
|---------|-----------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1      | Male            | Non-STEM         | Philosophy & History (6.8%); Languages & Literature (7.1%); <b>Mixed Engineering (5.6%)</b> ; Nursing & Healthcare (15.9%); Economics & Business (16.6%); Law & Political Sciences (9.8%)                                                       |
| C2      | Mixed           | Mixed            | Law & Political Sciences (10.8%); <b>Architecture &amp; Construction (3.6%)</b> ; Psychology & Therapy (4.0%); <b>ICT (7.2%)</b> ; Education (6.8%)                                                                                             |
| C3      | Male            | STEM             | Education (8.9%); Law & Political Sciences (11.8%); Economics & Business (15.9%); <b>Architecture &amp; Construction (4.0%)</b> ; <b>Mixed Engineering (4.3%)</b> ; <b>Arts &amp; Design (4.0%)</b> ; <b>Industrial Technologies (11.0%)</b>    |
| C4      | Female          | STEM             | <b>Biological Sciences (7.8%)</b> ; Communication & Documentation (5.6%); Mixed Social Sciences (5.0%); <b>Industrial Technologies (11.6%)</b> ; Psychology & Therapy (4.3%); <b>ICT (7.5%)</b> ; <b>Architecture &amp; Construction (3.4%)</b> |
| C5      | Female          | STEM             | Medicine & Biomedical Sciences (30.3%); <b>Experimental Sciences (9.8%)</b> ; <b>Industrial Technologies (12.3%)</b> ; Mixed Social Sciences (5.3%); Communication & Documentation (4.2%)                                                       |
| C6      | Mixed           | Mixed            | Psychology & Therapy (7.7%); Economics & Business (15.6%); <b>Architecture &amp; Construction (4.0%)</b> ; Education (7.7%); <b>Arts &amp; Design (3.8%)</b> ; Law & Political Sciences (10.1%); ICT (7.1%)                                     |
| C7      | Female          | STEM             | <b>Biological Sciences (11.1%)</b> ; <b>Experimental Sciences (5.6%)</b> ; Medicine & Biomedical Sciences (7.6%); Mixed Social Sciences (5.8%); <b>Industrial Technologies (12.2%)</b> ; Communication & Documentation (5.0%)                   |
| C8      | Female          | Non-STEM         | Nursing & Healthcare (26.0%); Social Intervention (8.1%); Education (9.6%); <b>Mixed Engineering (3.7%)</b> ; Psychology & Therapy (3.6%)                                                                                                       |

Note: STEM orientation reflects significant overrepresentation ( $p < .05$ ) of the aggregate STEM/non-STEM variable within each cluster; "Mixed" indicates no clear aggregate orientation. The final column reports only positively and significantly associated subdisciplines ( $p < .05$ ), with percentages indicating within-cluster shares. Percentages do not sum to 100 because only selected associated subdisciplines are displayed. STEM subdisciplines are highlighted in bold.

In contrast, male students are distributed across a broader range of clusters. Engineering-related disciplines and Industrial Technologies are present not only in high-performing profiles but also in intermediate and lower academic positions (Clusters 3 and 1). Even in Cluster 1, characterised by lower admission grades, Mixed Engineering maintains a visible presence. Cluster 3, a male academic-track profile concentrated in the middle deciles, is associated with Industrial Technologies, Architecture and Construction, and Mixed Engineering, alongside several non-STEM fields. This suggests that technical STEM fields remain accessible to men across a wider academic spectrum, rather than being confined to elite academic profiles.

Clusters with mixed gender composition (Clusters 2 and 6) display heterogeneous disciplinary orientations that reflect the broader diversity of the disciplinary space. Although both clusters

include some STEM-associated subdisciplines, such as ICT and Architecture and Construction, they also show substantial representation of Law and Political Sciences, Psychology and Therapy, Education, and Economics and Business. This indicates diversified academic profiles rather than clearly STEM-dominant orientations. Notably, subdisciplines such as Architecture and Construction, ICT, and Arts and Design appear within these intermediate clusters, functioning as hinge fields that connect technical, creative, and professional domains. These fields occupy intermediate positions within the disciplinary space and illustrate how certain programmes blur the boundaries between STEM and non-STEM areas. In this context, the clusters show that intermediate university admission grades do not systematically translate into STEM concentration, and that gender composition interacts with disciplinary allocation even among students with comparable grades.

Finally, Cluster 8 provides a marked contrast. Composed primarily of women entering through upper VET routes, with lower admission grades and employment commitments, this cluster is strongly associated with Nursing, Education, and Social Intervention. At the aggregate level, this cluster is significantly associated with non-STEM fields, while STEM subdisciplines remain marginal. This pattern indicates that vocational or non-linear access routes do not operate as meaningful pathways into STEM for women.

Taken together, the combined projection of subdisciplines and clusters reveals a structurally asymmetric pattern of STEM enrolment. Women's participation in technical, experimental and biological STEM subfields is concentrated within highly selective academic profiles, whereas men access STEM from a wider range of academic positions and entry contexts. To further assess this interpretation, a supplementary multinomial logistic regression model was estimated, modelling subfield choice as a function of admission grade, gender and their interaction. The joint test of the interaction terms indicates that the interaction between admission grade and gender is statistically significant ( $\chi^2(10) = 190.94$ ,  $p < .001$ ). Predicted probabilities from this model, reported in the Supplementary Material, show that the relationship between admission grade and field choice varies substantially by gender and subfield. These supplementary results support the interpretation that women's participation in technical STEM fields is concentrated in more academically selective entry profiles, while men's participation extends across a broader range of academic positions.

These patterns also allow us to revisit explanations based on people- versus object-oriented orientations (Su & Rounds, 2015). While prior research suggests that gender differences in vocational interests contribute to differential participation across STEM subdisciplines, our findings indicate that access to object-oriented technical fields is not simply a matter of individual preference. Women do enter highly technical disciplines such as Industrial Technologies and ICT; however, this occurs predominantly within the most academically selective clusters. By contrast, men are present in these same object-oriented fields across a much broader range of academic positions, including mid- and lower-performing profiles.

This asymmetry suggests that explanations based on gendered interest orientations remain incomplete unless situated within differentiated opportunity structures. Horizontal differentiation across subdisciplines is not merely a matter of individual preference, but is profoundly conditioned by the stratified architecture of university entry, where academic capital and access routes condition participation in ways that vary systematically by gender.

## *Discussion and conclusion*

This study shows that gendered participation in STEM is structured at the point of entry into higher education. Women's participation in technical, experimental and biological STEM subfields is concentrated in the most academically selective profiles, whereas men access STEM from a broader range of academic positions and entry contexts.

The contribution of the study can be summarised in relation to the three research questions. First, regarding the main dimensions structuring university entry, the results show that access to fields of study is organised primarily around a contrast between linear academic trajectories and vocational or non-traditional pathways. This extends research on horizontal inequalities in higher education (Charles & Bradley, 2009; Reimer & Pollak, 2010) by showing that field differentiation is embedded not only in social origin and gender, but also in access routes, age and employment conditions. Second, regarding student profiles, the cluster analysis shows that gender becomes salient within specific configurations of academic and social resources rather than operating as an isolated organising principle. This supports intersectional approaches (Gross et al., 2016; Nichols & Stahl, 2019) that emphasise how gender combines with educational trajectory, academic positioning and life-course conditions. Third, regarding STEM subfields, women's participation in technical, experimental and biological areas is concentrated in highly selective academic profiles, whereas men are present across a broader range of academic positions. This complements research on the internal heterogeneity of STEM (Cheryan et al., 2017; Su & Rounds, 2015) by showing that differences between subfields are also structurally anchored in differentiated entry contexts.

The results also confirm that STEM is internally heterogeneous. In particular, health-related fields occupy markedly different positions despite being treated as non-STEM in the operational classification adopted here. Medicine and Biomedical Sciences display an entry profile close to several STEM disciplines in terms of academic selectivity and linear entry pathways, whereas Nursing and Healthcare is associated more closely with vocational access, employment commitments and less academically selective profiles. This distinction reinforces the value of analysing subfields separately rather than relying exclusively on broad STEM/non-STEM categories.

These findings also qualify the *leaky pipeline* metaphor. Although recent research has advocated a more dynamic and non-linear pathways perspective (Li et al., 2025), the present results indicate that such trajectories are not gender-symmetric. Men enter STEM from a wider range of structural positions, whereas women's participation is concentrated within the most academically selective segments associated with linear trajectories. Gender inequalities in STEM therefore do not emerge only during university progression but are already embedded in differentiated conditions of access.

The results further refine explanations based on people- versus object-oriented interests (Su & Rounds, 2015) and gendered stereotypes associated with particular fields (Cheryan et al., 2017). Women do enter highly technical and object-oriented disciplines such as Industrial Technologies and ICT, but predominantly from the most academically selective profiles, whereas men are present across a broader spectrum of academic positions. Interest-based and cultural explanations therefore remain incomplete unless the structural conditions of access are also considered. The intermediate positioning of fields such as Architecture and Arts & Design

further illustrates that disciplinary boundaries are not always neatly divided between technical and non-technical domains.

These findings have implications for both research and policy. Future research should avoid treating STEM as a homogeneous category and examine how specific subfields are connected to different entry profiles. Longitudinal analyses are also needed to assess whether the entry patterns identified here translate into persistence, switching, dropout or later labour-market trajectories, while further attention should be given to non-enrolled applicants, alternative access routes and students with previous university enrolment. For policy, interventions aimed at reducing gender disparities should address not only interests and aspirations but also the stratified conditions shaping access to particular STEM subfields. In particular, strengthening pathways from vocational and non-linear routes into technical STEM fields may broaden opportunities for women beyond fields related to health, care and education.

Despite its contributions, this study has several limitations. First, the analysis focuses exclusively on first-time university entry among students accessing higher education through the two predominant admission routes and does not examine persistence, switching or dropout. Accordingly, the findings should be interpreted as applying to first-time entrants through these routes and should not be generalised to non-enrolled applicants, students entering through alternative routes or those with prior university enrolment. Second, although administrative data provide comprehensive coverage of the Catalan university system, they do not include subjective dimensions such as expectations, sense of belonging or prior socialisation experiences, which may also influence field-of-study choice. Third, the results depend on the operational definition of STEM adopted in the study. Health-related degrees are treated as a separate non-STEM area, although alternative definitions may classify some fields, particularly Medicine and Biomedical Sciences, differently. The findings should therefore be interpreted in relation to the classification used here. Importantly, the subfield-level analysis makes it possible to distinguish Medicine and Biomedical Sciences from Nursing and Healthcare and to show their clearly different positions within the university-entry space.

Taken together, the study shows that gender inequalities in STEM cannot be reduced to individual aspirations or preferences. Rather, they emerge at the intersection of gender-differentiated educational trajectories and stratified institutional opportunity structures. Situating gender within these contexts allows for a more structural understanding of horizontal segregation in higher education.

## Funding

No funding was received for this research.

## Disclosure statement

The authors declare no competing interests.

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